

WILP, BITS Pilani

MACHINE LEARNING · COMPANION READER

Decision Tree Classifier & Information Theory

Hunt's Algorithm, Entropy, Information Gain, and ID3 Construction

Session 7 · Read alongside the lecture slides · Every slide example worked out in full

Prepared for the Work Integrated Learning Programmes (WILP), BITS Pilani.

1 First Taste: Twenty Questions for Data

A decision tree splits complex decisions into a sequence of simple yes/no questions.

A **decision tree** converts tabular data into a flowchart structure. Each internal node tests an attribute, each branch represents an outcome, and each leaf node holds a class prediction.

The intuition

A decision tree breaks down a complex classification problem into a hierarchy of simple feature tests.

Decision Tree for PlayTennis Classification

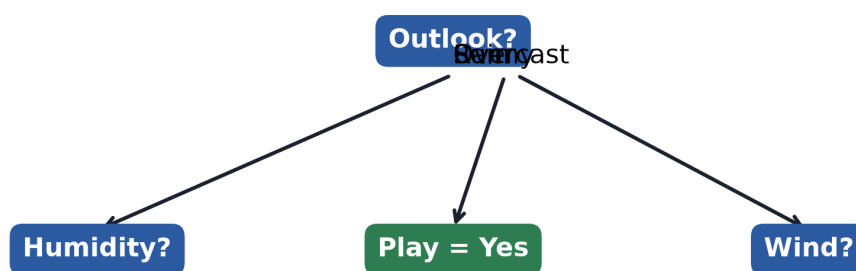


Figure 1: Structure of a decision tree showing decision nodes, feature branches, and pure leaf node predictions.

2 Decision Tree Representation & Expressiveness

Decision trees express disjunctions of conjunctions on attribute constraints.

Any decision tree can be rewritten as a set of logical rules.

2.1 Hunt's Algorithm

Given dataset D_t at node t :

1. If all records in D_t belong to class y_k , node t becomes a leaf labeled y_k .
2. If D_t contains multiple classes, select an attribute test condition to partition D_t into child subsets and recurse.

3 Information Theory & Entropy

Entropy measures the amount of impurity, disorder, or unexpected surprise in a dataset.

$$H(S) = - \sum_{i=1}^c p_i \log_2(p_i). \quad (1)$$

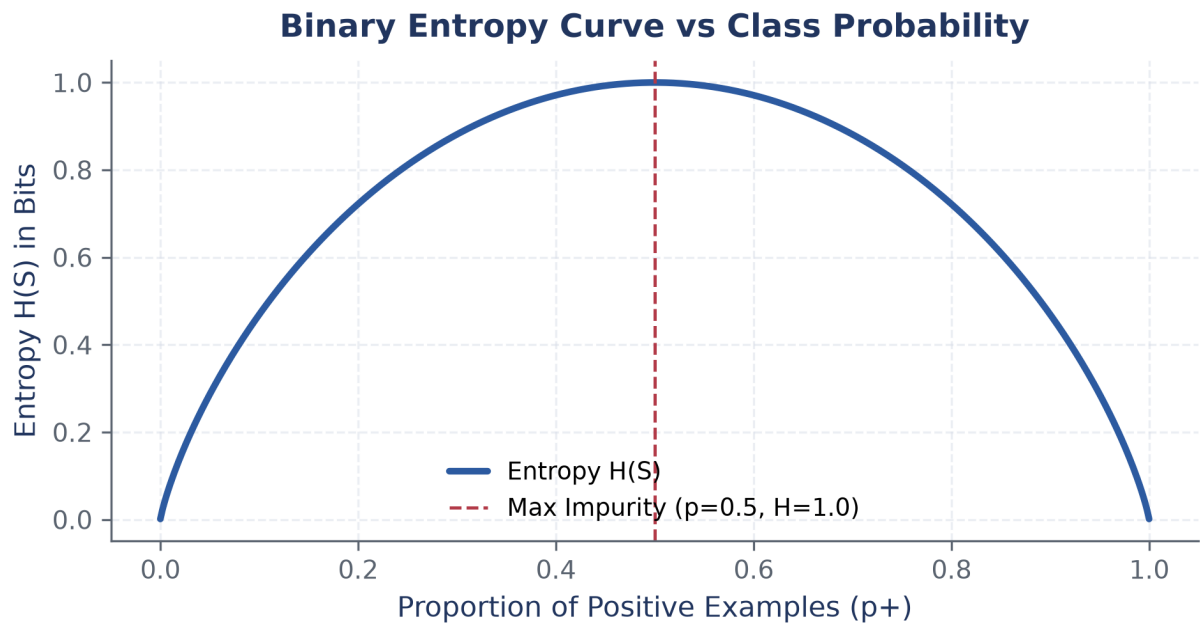


Figure 2: Binary entropy curve $H(S)$ as a function of positive class proportion p_+ .

Worked example: Computing Entropy for Sample Class Distributions

Calculate binary entropy for sample subsets:

Step 1. Pure dataset $[14+, 0-]$: $H(S) = 0$ bits.

Step 2. Balanced dataset $[7+, 7-]$: $H(S) = 1.0$ bit.

Step 3. Slightly imbalanced dataset $[9+, 5-]$: $H(S) \approx 0.940$ bits.

4 Information Gain & The ID3 Algorithm

Information Gain measures the expected reduction in entropy from splitting on an attribute.

$$\text{Gain}(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v). \quad (2)$$

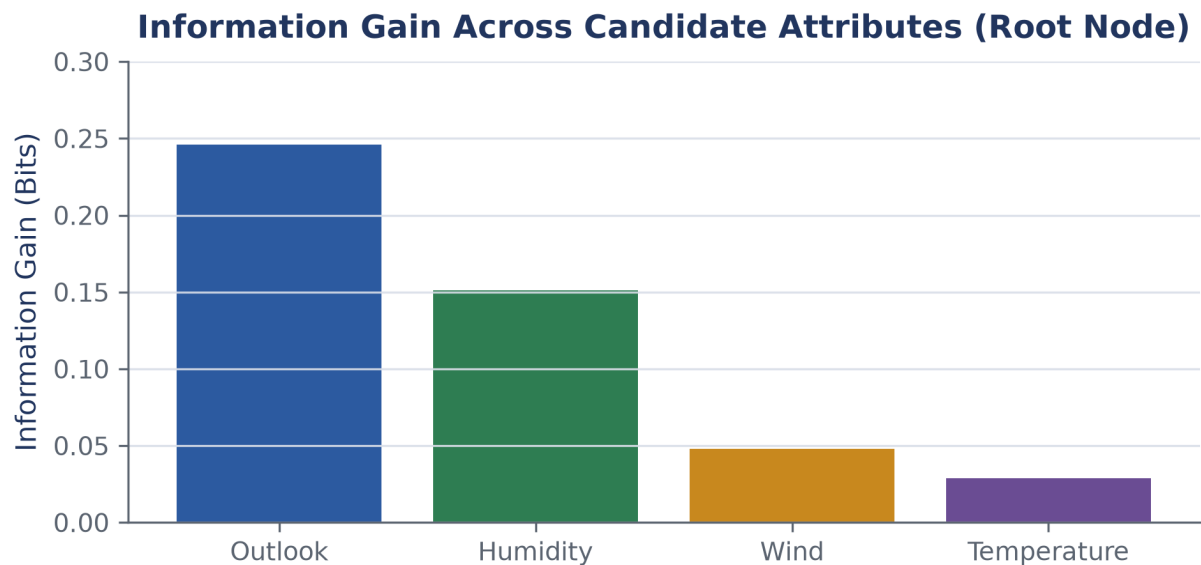


Figure 3: Information Gain values for candidate attributes at the root node.

Worked example: 14-Sample PlayTennis Root Split Selection

In the PlayTennis dataset ($S = [9+, 5-]$, $H(S) = 0.940$ bits):

Step 1. Evaluate Outlook: $\text{Gain}(S, \text{Outlook}) = 0.940 - 0.694 = 0.246$ bits.

Step 2. Compare Root Candidates: Outlook (0.246), Humidity (0.151), Wind (0.048), Temp (0.029). Outlook is chosen as root.